# World Models: Teaching AI to Imagine

ARTIFICIAL INTELLIGENCE · MACHINE LEARNING · DEEP LEARNING

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SWIPE ->

Most AI systems just react to their environment. World models go a step further - they learn an internal simulation of how the world works.

## What exactly is a world model?

A world model learns the dynamics of an environment. Given the current state and an action, it predicts what comes next.

Why this matters:

- Planning without acting: An agent can "imagine" future outcomes before taking real actions.
- Sample efficiency: Far fewer real interactions needed compared to pure trial-and-error.
- Safer decision making: Test risky actions inside a learned simulation first.
- Better generalization: Learn reusable knowledge about physics, cause and effect.

The idea isn't new. "World Models" by Ha & Schmidhuber (2018) showed agents could learn inside their own dreams. But it's seeing a resurgence now.

Modern approaches like JEPA (Joint Embedding Predictive Architecture), V-JEPA, and model-based RL agents like DreamerV3 are taking this further.

Instead of predicting every pixel, they predict in abstract representation space - focusing on what actually matters.

The key insight:

Don't teach AI to watch the world frame by frame. Teach it to understand how the world evolves.

That's the path towards more capable, efficient and grounded agents.

#WorldModels #JEPA #ModelBasedRL #RepresentationLearning #SelfSupervisedLearning #AI #DeepLearning #MachineLearning

# Thanks for reading.

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Code: github.com/yasirusman85/Tank